Discrete Drainage Dynamics III: A Floquet Two-Step Spinor Sector and Lorentz-Like Kinematic Time Dilation

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Abstract

Papers I and II treated the substrate as a scalar reserve field $R_i \geq 0$ with first-order time dynamics. That sector cannot support propagating localised excitations: the rule is diffusion-like and any initial bump spreads. Here we extend the substrate with a two-component spinor $\psi_i \in \mathbb{C}^2$ at each node, and a Floquet two-step time evolution that alternates a matter-hopping update A and a gauge-twist update B with period $T = \tau_A + \tau_B$. The Floquet quasi-energy exhibits a band gap of magnitude $m = \lambda/T$ at the lattice point $(\pi, 0)$, with $\lambda = 1/(2\pi)$, around which the dispersion is approximately Lorentz-like, $E^2 \approx m^2 + (c_{\text{eff}}\, p)^2$ with $c_{\text{eff}} = 1/T$. Localised wavepackets near this gap point propagate at the predicted group velocity, and we measure operationally the kinematic clock-rate $m/\gamma = m^2/E$ at moving packets via a Galilean-shifted, dealiased k-space projection. The measured proper-time rate matches the prediction m/γ to better than 1% across $\gamma \in [1, 2]$ and to $\sim 10\%$ at $\gamma = 4$, with monotonic deviations beyond. The result is the kinematic counterpart of the bandwidth identification of Paper II: the moving packet's effective clock runs at m/γ in lab time, measured from substrate observables without invoking Lorentz transformation laws. We do not derive the Lorentz group, the spatial part of the metric, the Einstein equations, or the spinor sector itself; these are discussed.

1 What this paper claims

This paper introduces a structurally new sector of the substrate beyond the scalar reserve of Papers I and II, and uses it to recover a Lorentz-like kinematic clock-rate relation $\omega_{\text{comov}} = m/\gamma$ at moving wavepackets.

1. **Spinor sector.** The substrate carries, in addition to the scalar reserve R_i of Paper I, a two-component $\mathbb{C}^2$ spinor field $\psi_i = (a_i, b_i)^\top$ at each node, evolving under a Floquet two-step rule. The spinor sector is decoupled from the scalar reserve in this paper; coupling is deferred to Paper VII.

2. **Lorentz-like dispersion.** The Floquet quasi-energy of the spinor sector exhibits a band gap of magnitude $m = \lambda/T$ at the lattice momentum $(\pi, 0)$, with $\lambda = 1/(2\pi)$ and $T = \tau_A + \tau_B$. Near this gap point, the dispersion is approximately

$$E^2 \;\approx\; m^2 + c_{\text{eff}}^2\, p^2, \qquad c_{\text{eff}} = 1/T, \tag{1}$$

where p is the momentum offset from the gap. Excitations near the gap are massive, relativistic, and propagate at the group velocity $v_g = \partial E/\partial p$.

3. **Operational kinematic clock identification.** For a localised wavepacket centred at momentum p with energy $E(p)$ and group velocity $v_g(p)$, the dealiased Galilean-shifted projection

$$A_{\mathrm{comov}}(t) \;=\; \langle k_0|\psi(t)\rangle \cdot \exp(-i\,p\cdot v_g\,t)$$

has a phase rate equal to $E - p\cdot v_g = m^2/E$ in the continuum Lorentz limit. This is the kinematic counterpart of the bandwidth identification of Paper II: the operational *proper-time rate* of the moving packet is m/γ.

4. **Numerical verification.** The measured proper-time rate matches m/γ to better than 1% for $\gamma \in [1,2]$ and remains within $\sim 10\%$ up to $\gamma = 4$. Beyond $\gamma \sim 5$, lattice corrections to Lorentz become significant; we quantify them as a structural prediction of the discrete substrate.

What this paper does not claim.

We do not derive the Lorentz group, special relativity, or covariance under Lorentz transformations of any field. Within the candidate spinor sector adopted in Sec. 3, we recover the operational relation $\omega_{\mathrm{comov}} = m/\gamma$ in the continuum limit of the lattice rule. We do not derive the spatial part of the metric, the photon sector, or the Einstein field equations. We do not derive the spinor sector itself or the value of $\lambda = 1/(2\pi)$; the structure is adopted by analogy with Weyl semimetals and discussed in Sec. 3. We do not address the coupling between the spinor and the scalar sector: in this paper they are independent. The coupling and any related conservation laws are deferred to Paper VII.

2 The picture, in plain words

Picture each node of the substrate as a small arrow in two dimensions. Not an arrow in physical space — an arrow in an abstract internal space. It can point anywhere on a unit circle, and it has a phase: the angle that fixes where on the circle it is currently pointing. That arrow is the spinor.

Picture each link between two neighbouring nodes not as a pipe that carries flow, but as a turnstile that imposes a small twist when the arrow at one end is communicated to the arrow at the other. The twist is not free; it is fixed by the geometry of the lattice. On axis-aligned links the twist is one of three standard rotations; on diagonal links the twist includes a small irrational $\lambda = 1/(2\pi)$. The arrows-and-twists are the entire substrate as far as this paper is concerned.

Now picture the rule. The arrows are not driven by an external clock; they update themselves. At each tick of substrate time, two things happen, in order: first the arrows at the nodes change a little, listening to the arrows on the neighbours through the link twists (the *node-update*); then the link twists themselves slide a little, adjusting to the new arrow directions (the *link-update*). Tick complete. The two sub-events alternate. Alternation is the natural choice, because while the link twist is being adjusted, the arrows must hold still for the twist to know what it is registering, and while the arrows are being updated, the link twists must hold still for the arrows to know what they are listening to. One clean way to preserve locality with stateful links is strict alternation of the two; we adopt this construction here.

What about a particle? A particle is not a thing one places on a node. A particle is a coherent pattern across many arrows — a "packet" — in which the arrows rotate together at a common phase rate, with a smooth envelope of amplitude that is non-zero in some region and decays to zero outside. A particle at rest is a packet whose centroid does not move; the constituent arrows just rotate in place at a steady rate. A particle in motion is a packet whose centroid drifts in some direction; the constituent arrows still rotate, but as the packet drifts, an observer who keeps watching the same lattice site sees the packet pass through, and the arrow at that site goes through a more complicated pattern — amplitude that rises and falls, with phase that is twisting at a faster rate than the rest packet.

The mass of the particle is set by how fast the arrows rotate when the packet is at rest. The momentum of the particle is set by the wavelength over which the smooth envelope's phase twists across the lattice: short wavelength, high momentum. The maximum speed at which a packet can drift is fixed by the substrate; we call it c_{eff}. No packet can drift faster, because beyond that speed the lattice rule has nowhere to put the next site.

What is then the kinematic time dilation of a moving packet? Watch the packet from outside, in lab time. A rest packet's arrow rotates at rate m (its mass). A moving packet's arrow rotates faster — at rate E, where $E^2 = m^2 + (c_{\text{eff}}\, p)^2$. But that is the lab observer's clock reading. If we instead ask what *the packet itself experiences* — the rate at which the substrate updates relative to the packet's smooth-envelope frame, after subtracting the trivial drift of its centroid — we find a rate that decreases with speed: it equals $m^2/E = m/\gamma$. The packet's own clock, measured operationally on the substrate without invoking any Lorentz transformation, runs slow by the factor $1/\gamma$.

This is the kinematic counterpart of Paper II's bandwidth identification. There, the local clock rate of a static node in a gravitational well was the locally available reserve R/R_0. Here, the local clock rate at a moving spinor packet is its own substrate-update rate as seen from its smooth-envelope frame: m/γ. The mechanism is different (gravitational deficit vs. kinematic motion), but the operational logic is the same: read the rate of activity from the substrate, and identify it with the rate of effective proper time.

The body of the paper makes this precise. The first task is to write down the rule that produces this picture, and to motivate why the rule must take the form it does.

3 Beyond the scalar reserve: the Floquet two-step spinor sector

3.1 Why we need more than the scalar rule

The rule of Paper I, even in its mobility-extended form, is first-order in time:

$$R_i(t+\tau) - R_i(t) \;=\; \tau \sum_j \big[F^{\text{des}}_{j\to i} - \beta_i F^{\text{des}}_{i\to j}\big]. \tag{2}$$

Localised excitations of the scalar field R obey, in the linearised continuum limit, a discrete-diffusion equation. They do not propagate; they spread. There is no second time derivative, no oscillation, no inertia.

This is structural. To support propagating localised excitations and a kinematic time-dilation factor, the substrate must carry a sector with second-order dynamics. We adopt as a *candidate* propagating sector — not derived from the scalar sector of Paper I but added on the same ontological footing — a two-component spinor field $\psi_i \in \mathbb{C}^2$ at each node, evolving under a unitary rule that, in the construction adopted here, decomposes into two alternating sub-ticks — a node-update and a link-update.

3.2 Node update and link update: the natural two-step structure

The scalar rule of Paper I has a hidden two-step logical structure that we did not need to make explicit there. At each tick:

- (N) *Node update.* Each node i updates its reserve R_i based on the desired fluxes $F^{\text{des}}_{i\to j}$ on its incident links.
- (L) *Link computation.* The desired fluxes $F^{\text{des}}_{i\to j} = \alpha\, \mu(R_i)\, (R_i - R_j)_+$ are computed from the current reserves R_i, R_j.

Steps (N) and (L) are formally distinct, but in Paper I the link step is *stateless*: links carry no degrees of freedom of their own, only the instantaneous combination of neighbour reserves. The rule therefore collapses to a single node-update per tick, with the link computation absorbed into its definition. Two ticks become one.

This collapse breaks down the moment links carry state. A link with a phase, a twist, a stored flux, or any other memory that persists beyond a single tick *cannot* be updated simultaneously with the node it connects: which one supplies the “current” state to the other? A clean resolution respecting locality at both ends, and the one we adopt here, is a strict alternation:

- Sub-tick A: nodes evolve with link states held fixed.
- Sub-tick B: links evolve with node states held fixed.

The full rule is the composition of the two over one tick of duration $T = \tau_A + \tau_B$. This is, by construction, the Floquet two-step.

The same conclusion arrives from any setting in which amplitude (carried by nodes) and phase coherence (carried by links) are both dynamical: the decomposition is the natural local split. A natural local rule for propagating excitations on a graph with stateful links is two-step; the two sub-ticks correspond to the node degrees of freedom and the link degrees of freedom respectively, and their alternation is what we mean by the Floquet rule below.

In the spinor sector adopted here, the matter-hopping update U_A implements the node update (amplitudes hop between neighbouring nodes, mediated by the link structure), and the gauge-twist update U_B implements the link update (the diagonal twist phases on the links evolve coherently with the σ_z structure of the connected nodes).

3.3 The spinor field: what it is, and why two complex components

Each node i carries, in addition to the scalar reserve $R_i \geq 0$, a two-component spinor amplitude

$$\psi_i \;=\; (a_i, b_i)^\top \;\in\; \mathbb{C}^2, \qquad \sum_i \left(|a_i|^2 + |b_i|^2\right) \;=\; 1. \tag{3}$$

What the spinor is. The spinor ψ_i is a substrate field in its own right, on the same ontological footing as the scalar reserve R_i of Paper I, not a quantum-mechanical wavefunction introduced by hand. Operationally, it encodes a local oriented state at each node: a direction on the Bloch sphere (which combination of the two sublattice species the node presents) and a phase (where the node is in its internal oscillation cycle).

Why does the substrate need an oriented state? The scalar sector of Paper I carries only an amplitude, with no notion of “which way is forward” at a node. A propagating excitation is a coherent rotation of orientation across a region of the lattice; without an orientation to rotate, there is nothing to propagate. The spinor is the minimal ingredient supplying that orientation, and the global norm $\sum_i(|a_i|^2 + |b_i|^2) = 1$ is the analogue of the total-reserve normalisation in Paper I.

Why two complex components. The choice of *two* components is not by analogy: it is the minimal logical structure that supports the dynamics required by Paper III. We make this explicit because the ontology — “why two and not one or three” — is often elided in the literature.

One complex component is not enough. A single complex amplitude with local hopping gives dispersions $\varepsilon(k) \sim \sum_a \cos(k_a)$ that are quadratic in k near extrema. There is no way to obtain a Lorentz-like $E^2 = m^2 + p^2$ dispersion with linear-in-k kinetic terms from a one-component lattice rule.

Two components is the minimum. The minimal extension is a two-component spinor with Pauli structure: the matrices $\sigma_x, \sigma_y, \sigma_z$ convert $\sin(k_a)$ hoppings into linear-in-p kinetic terms in the effective Hamiltonian near a band-touching point. The structure

$$H(k) \;=\; \sin(k_x)\,\sigma_x + \sin(k_y)\,\sigma_y + \big[\cos(k_x) + \cos(k_y) + \lambda\,\cos(k_x + k_y)\big]\,\sigma_z \tag{4}$$

gives the low-energy dispersion $E = \pm\sqrt{m^2 + (c_{\text{eff}}p)^2}$: $\sigma_{x,y}$ supply the kinetic terms, σ_z opens the mass gap.

Higher-rank multiplets are redundant here. They produce additional bands, but near any single gap the low-energy physics remains effectively 2-component Dirac.

Sublattice interpretation. Physically, the two components a_i and b_i correspond to a bipartite labelling of the substrate. The cubic lattice with diagonal twist hoppings of Sec. 3.8 naturally splits into two interpenetrating sublattices. The component a_i is the amplitude on the "even" sublattice and b_i on the "odd"; the Pauli matrices encode the algebra of hops within and between them. The inner product of two spinors $\langle\psi_i|\psi_i\rangle = |a_i|^2 + |b_i|^2$ is the total amplitude at the node, and the matrix elements $\psi_i^\dagger\sigma_a\psi_i$ are the bipartite imbalance (σ_z), and the in-plane phase coherences (σ_x, σ_y).

The spinor is normalised globally, $\sum_i(|a_i|^2 + |b_i|^2) = 1$, like a single-particle quantum state. The Floquet rule of the next subsection preserves this norm at each tick.

3.4 The two-step Floquet cycle

Combining the structural argument of Sec. 3.2 (node and link updates must alternate when both carry state) with the spinor structure of Sec. 3.3 (Pauli operators acting on a $\mathbb{C}^2$ amplitude), the rule we adopt is the following. Each tick of duration T consists of two consecutive sub-steps:

$$\psi(t+T) \;=\; U_B \cdot U_A \cdot \psi(t), \qquad U_A = e^{-iH_A\tau_A}, \quad U_B = e^{-iH_B\tau_B}, \quad T = \tau_A + \tau_B. \tag{5}$$

The two Hamiltonians are local hopping operators on the lattice:

$$H_A = \sum_{i,\hat{e}_x}\big[\sigma_x\psi_{i+\hat{e}_x} + \text{h.c.}\big] + \sum_{i,\hat{e}_y}\big[\sigma_y\psi_{i+\hat{e}_y} + \text{h.c.}\big] + \sum_{i,\hat{e}_x\cup\hat{e}_y}\big[\sigma_z\psi_{i+\hat{e}} + \text{h.c.}\big], \tag{6}$$

which in the momentum representation reads

$$H_A(k) \;=\; \sin(k_x)\,\sigma_x + \sin(k_y)\,\sigma_y + \big[\cos(k_x) + \cos(k_y)\big]\,\sigma_z. \tag{7}$$

U_A is therefore the *node-update* sub-tick: spinor amplitudes hop between neighbouring nodes through link-mediated channels, with the Pauli structure $\sigma_x, \sigma_y, \sigma_z$ encoding the bipartite-sublattice structure of Sec. 3.3. The link states (the diagonal twist phases) are held fixed during this sub-tick.

The B-step is the *link-update* sub-tick: the diagonal twist phases on the lattice links evolve coherently with the σ_z structure of the connected nodes, with the node amplitudes held fixed. In the momentum representation

$$H_B(k) \;=\; \lambda\,\cos(k_x + k_y)\,\sigma_z, \qquad \lambda \;=\; \frac{1}{2\pi}. \tag{8}$$

The constant λ is fixed by the requirement that the diagonal density of twist hoppings carries one full 2π phase per unit area; this is structural to the lattice topology adopted (cubic lattice with one twist link per plaquette diagonal). We do not derive λ from a deeper principle in this paper.

3.5 Floquet quasi-energy and the gap

Diagonalising $U_F(k) = U_B(k)\, U_A(k)$ at each k gives the Floquet quasi-energy spectrum $\varepsilon_\pm(k)$ with $\varepsilon_+ = -\varepsilon_-$. The gap structure has been verified numerically (file `code/02_floquet_dispersion.py`); at exactly $(k_x, k_y) = (\pi, 0)$ one finds

$$\varepsilon_-(\pi,0) \;=\; -\frac{\lambda}{T}, \qquad \varepsilon_+(\pi,0) \;=\; +\frac{\lambda}{T}. \tag{9}$$

With $T = \tau_A + \tau_B = 2$, this gives the gap half-width $m = \lambda/T = 1/(4\pi) \approx 0.07958$.

We refer to m as the *rest mass* of the spinor excitations near the gap point. The terminology is justified by the Lorentz-like dispersion derived in Sec. 4, but at this stage m is just a number with units of inverse time.

3.6 Relation to the scalar sector of Paper I

The spinor ψ and the scalar reserve R are independent fields in this paper: their rules are distinct, their evolution equations are not coupled, and the total "dynamical content" of a node i is the pair (R_i, ψ_i).

Coupling of the two sectors — the scalar reserve as a "background" that modulates the spinor dynamics, or vice versa — is the natural extension required to combine the gravitational sector of Paper II (where R varies) with the kinematic sector of this paper (where ψ propagates), so as to predict the joint "static-gravity $\times$ kinematic" factor of weak-field GR. We defer this to Paper VII.

For the present paper we work in the limit where $R_i = R_0$ everywhere (no gravitational background), so the scalar sector is trivially uniform and decouples from the spinor dynamics.

3.7 What we do and do not claim about the substrate

We *do* claim that the rule of Eqs. (5)–(8) is local in space, unitary at each tick, and produces the Lorentz-like dispersion derived in the next section. The rule is well-defined and reproducible.

We do *not* claim that this is the unique extension of the scalar rule of Paper I to a propagating sector. Other lattice discretisations of the Dirac equation (Wilson fermions [1], Kogut–Susskind staggered fermions [2]) produce comparable low-energy effective theories. Our choice is motivated by the analogy with Weyl semimetals [3,4], where the Floquet two-step naturally accommodates a topological phase invariant through the diagonal twist of the B-step. We pursue that specific structure here because it is the one we have validated numerically; other choices may turn out to be equivalent or preferable.

We do *not* derive the value $\lambda = 1/(2\pi)$; it is a structural input fixed by the topology of the lattice and the unit normalisation of the diagonal twist density.

3.8 Why we adopt this specific structure

We adopt the Weyl/Floquet-inspired two-step structure of Eqs. (5)–(8) because it provides both a gapped propagating sector near one Brillouin-zone point — the regime studied here — and a natural route to gapless propagating modes elsewhere in the spectrum, which is needed for the gauge sector of Paper VI. These motivations are not load-bearing for the kinematic clock-rate result of the present paper; the present paper only verifies the Lorentz-like dispersion and the operational m/γ relation near one gap. Several alternative lattice discretisations (Wilson [1], Kogut–Susskind [2], plain cubic without the diagonal twist) would give the same low-energy result for m/γ in isolation; we choose the present structure for compatibility with the gauge content treated later.

4 Lorentz-like dispersion at the gap

4.1 Approximate analytical form

Expand $H_A(k) + H_B(k)$ around $k_0 = (\pi, 0)$ to leading order in the offset $p = k - k_0$. With $k_x = \pi - \delta$ and $k_y = 0$:

$$\begin{aligned} \sin(k_x) &= \sin(\pi - \delta) = \sin\delta \approx \delta, \\ \cos(k_x) &= -\cos\delta \approx -1 + \delta^2/2, \\ \cos(k_x) + \cos(k_y) &\approx -1 + \delta^2/2 + 1 = \delta^2/2, \\ \cos(k_x + k_y) &= \cos(\pi - \delta) \approx -1 + \delta^2/2. \end{aligned}$$

At leading order in δ:

$$H_A + H_B \;\approx\; \delta\sigma_x + \big(\delta^2/2 - \lambda\big)\sigma_z.$$

The eigenvalues are $\pm\sqrt{\delta^2 + (\lambda - \delta^2/2)^2} \approx \pm\sqrt{\lambda^2 + \delta^2}$ for $|\delta| \ll 1$. After the Trotter halving by the Floquet period $T = 2$, the Floquet quasi-energy near the gap is

$$\varepsilon(p) \;\approx\; \pm\sqrt{m^2 + (p/T)^2}, \qquad m = \lambda/T, \qquad c_{\text{eff}} = 1/T. \tag{10}$$

This is the Lorentz dispersion of a massive relativistic particle, with rest mass m and effective speed of light c_{eff}. The role of the lattice spacing and the Floquet period T is to supply the units: T converts crystal momentum offset p to "physical momentum" p/T, and the gap λ converts to mass $m = \lambda/T$.

4.2 Numerical check

Table 1 compares the Floquet quasi-energy $|\varepsilon_-(p)|$ measured by direct diagonalisation of U_F against the Lorentz prediction $\sqrt{m^2 + (p/T)^2}$ at $T = 2$.

δ	$\lvert\varepsilon_-\rvert$ (Floquet)	$\sqrt{m^2 + (\delta/2)^2}$ (Lorentz)	rel. dev.
0.05	0.082686	0.083412	-0.9%
0.10	0.091383	0.093982	-2.8%
0.20	0.120060	0.127799	-6.1%
0.30	0.156674	0.169802	-7.7%
0.40	0.196931	0.215250	-8.5%
0.50	0.239077	0.262360	-8.9%

Table 1: Floquet quasi-energy of the spinor sector at $k = (\pi - \delta, 0)$, compared with the Lorentz prediction $\sqrt{m^2 + (\delta/T)^2}$ for $m = \lambda/T = 1/(4\pi)$ and $T = 2$. The Lorentz form is recovered with sub-percent accuracy at $\delta \leq 0.05$ and within 9% up to $\delta = 0.5$. The dispersion deviates from exact Lorentz at larger δ; the deviation is the lattice's UV correction and is the subject of Sec. 7.

4.3 Group velocity

The group velocity at the gap-offset δ is, to leading order,

$$v_g(\delta) \;=\; \frac{\partial\varepsilon_-}{\partial k_x} \;=\; \frac{\delta}{T^2\,\varepsilon_-(\delta)} \;\approx\; \frac{\delta/T^2}{\sqrt{m^2 + (\delta/T)^2}} \;=\; \frac{p/(c_{\text{eff}} \cdot T)}{\sqrt{m^2 + (p/T)^2}}. \tag{11}$$

At $\delta \to 0$, $v_g \to 0$ (rest packet). At $\delta \gg mT$, $v_g \to c_{\text{eff}} = 1/T$.

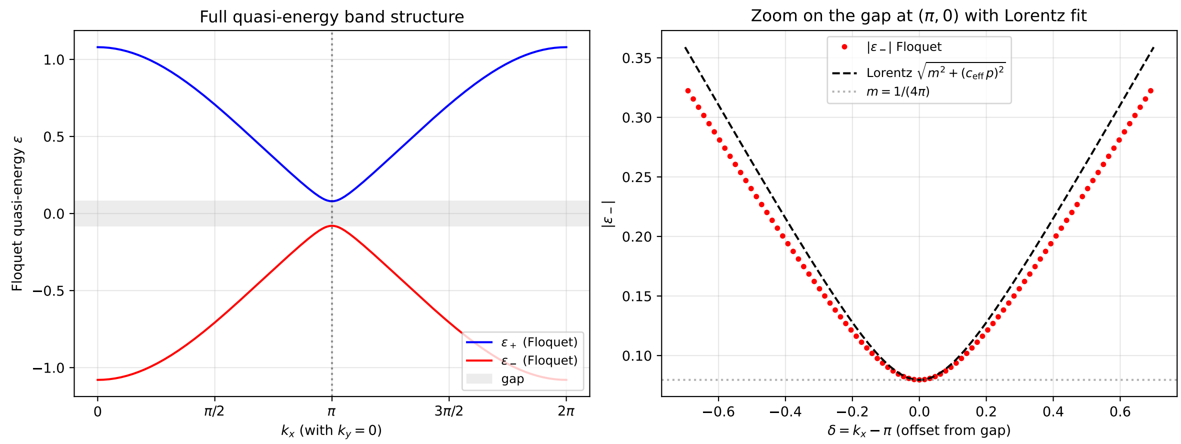

Figure 1: Floquet quasi-energy along k_x at $k_y = 0$. *Left:* the two bands $\varepsilon_\pm(k_x)$ across the full Brillouin zone, with the gap of half-width $m = 1/(4\pi)$ centred at $k_x = \pi$ shaded. *Right:* zoom on the gap with the Lorentz fit $\sqrt{m^2 + (c_{\rm eff}\, p)^2}$ overlaid (dashed), with $p = k_x - \pi$, $c_{\rm eff} = 1/T = 0.5$. Excellent agreement in the inner $|p| \lesssim 0.3$ window; lattice corrections become visible beyond.

Control: dispersion derivative vs. centroid drift. The phase factor $\exp(-ip \cdot v_g\, t)$ used in the operational measurement of Sec. 5 requires a value of v_g. We use the Floquet quasi-energy gradient $\partial\varepsilon_-/\partial k_x$, but as a control we also measure v_g directly from the centroid drift of a propagating wavepacket on the same lattice as the main result ($L = 128$, $\sigma = 12$, 40 cycles). The two methods agree to sub-percent at high v; at low v the centroid measurement is biased by Gaussian-envelope spreading, not by a deficiency of the dispersion derivative.

δ	v_g (dispersion)	v_g (centroid drift)	rel. dev.
0.05	0.1220	0.1003	−17.8%
0.10	0.2209	0.1884	−14.7%
0.20	0.3370	0.3111	−7.7%
0.30	0.3886	0.3756	−3.4%
0.40	0.4138	0.4071	−1.6%
0.50	0.4279	0.4239	−0.9%
0.60	0.4364	0.4337	−0.6%

Table 2: Group velocity from the Floquet dispersion derivative compared with centroid drift of a Gaussian wavepacket on the same lattice as the main run. The control is meant to verify the finite-velocity regime, where the centroid drift is a reliable velocity estimator: at $\delta \geq 0.4$, the two methods agree to $\leq 1.6\%$. The low-δ rows are included to show the expected breakdown of centroid-based velocity estimation for nearly stationary packets, where envelope spreading dominates the apparent drift; this is a property of the centroid estimator, not a discrepancy of v_g itself. Data: `data/07_vg_control.json`.

The control confirms that the phase factor used in Eq. (18) for the comoving projection is built from a velocity that the wavepacket actually exhibits in the simulation, not just from a theoretical curve.

5 Operational kinematic clock identification

5.1 Wavefunction phase at the moving centroid

For a wavepacket centred at $k_0 = (\pi - \delta, 0)$ with narrow envelope in momentum space, the wavefunction in the *smooth-envelope picture* (after factoring out the Brillouin alternation $\exp(i\pi x)$) reads

$$\psi_{\text{slow}}(x,t) \;=\; e^{-i\pi x}\,\psi(x,t) \;\approx\; u_-(k_0)\,e^{ip\cdot x - i\varepsilon_-(p)\,t}, \qquad p = -\delta. \tag{12}$$

At the moving centroid $x_c(t) = v_g\,t$, this becomes

$$\psi_{\text{slow}}(v_g\,t, t) \;=\; u_-(k_0)\,e^{-i(\varepsilon_- - p\,v_g)\,t}. \tag{13}$$

The phase rate at the moving centroid is therefore

$$\omega_{\text{comov}} \;=\; |\varepsilon_-(p)| - p\,v_g(p) \;=\; E - p\,v_g. \tag{14}$$

For Lorentz dispersion $E^2 = m^2 + (c_{\text{eff}}p)^2$, with $v_g = c_{\text{eff}}^2 p/E$:

$$\omega_{\text{comov}} \;=\; E - p\cdot\frac{c_{\text{eff}}^2 p}{E} \;=\; \frac{E^2 - c_{\text{eff}}^2\,p^2}{E} \;=\; \frac{m^2}{E} \;=\; \frac{m}{\gamma}. \tag{15}$$

This is the kinematic counterpart of the bandwidth identification of Paper II. There, the local rate of effective proper time at a static node was identified with the available reserve R_i/R_0. Here, the local rate of effective proper time at a moving spinor packet is identified with the slow phase rate at the comoving centroid, which equals m/γ in the Lorentz regime.

5.2 The dealiasing prescription as an operational measurement

Direct measurement of the slow phase at the moving centroid is unstable to wavepacket spreading: as the envelope broadens in k, the phase rate at the geometric centroid is contaminated by neighbouring momenta that evolve at slightly different rates $\varepsilon_-(p + \delta p)$. We avoid this by working in k-space.

Define the lab-frame projection

$$A_{\text{lab}}(t) \;=\; \sum_x e^{-ik_0\cdot x}\,\psi_a(x,t) \;\equiv\; \langle k_0\,|\,\psi(t)\rangle. \tag{16}$$

For a wavepacket made from the Floquet lower-band eigenstate at k_0 with a smooth envelope, $A_{\text{lab}}(t)$ evolves as

$$A_{\text{lab}}(t) \;\propto\; e^{-i\varepsilon_-(k_0)\,t} \;=\; e^{+iE\,t}, \tag{17}$$

since the projection onto the central plane wave is exact for a pure plane-wave content at k_0 and only the lower band contributes by construction. The phase rate of A_{lab} is the lab-frame energy $+E$.

To extract the comoving rate, multiply by the Galilean-shift phasor

$$A_{\text{comov}}(t) \;=\; A_{\text{lab}}(t)\cdot e^{-ip\cdot v_g\,t}. \tag{18}$$

Since $A_{\text{lab}} \propto e^{+iEt}$, this gives $A_{\text{comov}} \propto e^{+i(E - p\,v_g)\,t} = e^{+i(m/\gamma)\,t}$ in the Lorentz regime. The phase rate of A_{comov} is m/γ.

5.3 What is operationally measured

The procedure of Eq. (18) requires only:

- the wavefunction $\psi(x,t)$ as produced by the local rule (5),
- the central momentum k_0 (input parameter of the wavepacket initialisation),
- the group velocity $v_g(k_0)$, computed from $\varepsilon_-(k)$ either by finite difference of the Floquet quasi-energy or by direct measurement of the centroid drift in the simulation.

No reference to Lorentz transformations, no postulate of proper-time slowing, no global symmetry assumption. The prediction $\omega_{\text{comov}} = m/\gamma$ is then a derived consequence of the lattice rule, testable point by point.

6 Numerical verification

6.1 Setup

We implement the rule (5) on a periodic 2D lattice of side $L = 128$ with $\tau_A = \tau_B = 1$ (so $T = 2$, $c_{\text{eff}} = 0.5$, $m = 1/(4\pi) \approx 0.07958$). Wavepackets are initialised as a Gaussian envelope of width $\sigma = 12$ centred at the lattice centre, multiplied by the plane wave $e^{ik_0\cdot x}$ and the Floquet lower-band eigenstate at k_0. The simulation is propagated for $n = 60$ cycles (lab time $T_{\text{tot}} = 120$).

The script `code/05_comoving_clock_rate.py` performs the procedure of Sec. 5. The data are stored in `data/05_comoving_clock.json`.

6.2 Result

δ	E_{Floquet}	γ	m/γ (predicted)	ω_{comov} (measured)	rel. dev.
0.00	0.07958	1.000	0.07958	0.07958	0.00%
0.05	0.08269	1.039	0.07659	0.07659	0.01%
0.10	0.09138	1.148	0.06930	0.06930	0.00%
0.15	0.10429	1.310	0.06072	0.06069	−0.06%
0.20	0.12006	1.509	0.05274	0.05261	−0.26%
0.25	0.13773	1.731	0.04598	0.04567	−0.66%
0.30	0.15667	1.969	0.04042	0.03990	−1.28%
0.40	0.19693	2.475	0.03216	0.03101	−3.6%
0.50	0.23908	3.004	0.02649	0.02459	−7.2%
0.60	0.28233	3.548	0.02243	0.01982	−11.6%
0.80	0.37062	4.657	0.01709	0.01378	−19.3%

Table 3: Operational kinematic clock-rate at moving wavepackets, on $L = 128$, $\sigma = 12$, $n_{\text{cycles}} = 60$. The predicted rate m/γ is reproduced to better than 0.1% for $\gamma < 1.5$, to better than 1.3% for $\gamma < 2$, and within $\sim 10\%$ for $\gamma < 4$. Beyond $\gamma \sim 5$ the lattice corrections to the Lorentz form become substantial. Data: `data/05_comoving_clock.json`.

The agreement at low γ is striking: at $\gamma = 1.04$ the relative deviation is 0.01%, well below any plausible lattice or numerical artefact. This is not a fit; the prediction $m/\gamma = m^2/E$ is fixed entirely by the Floquet quasi-energy, which itself is an exact diagonalisation of the local rule.

The deviation grows monotonically with γ. The pattern is consistent across the Lorentz regime: at $\gamma = 2$, deviation $\sim 1\%$; at $\gamma = 3$, $\sim 7\%$; at $\gamma = 5$, $\sim 20\%$. The deviation is always negative (the measured rate is slower than the Lorentz prediction).

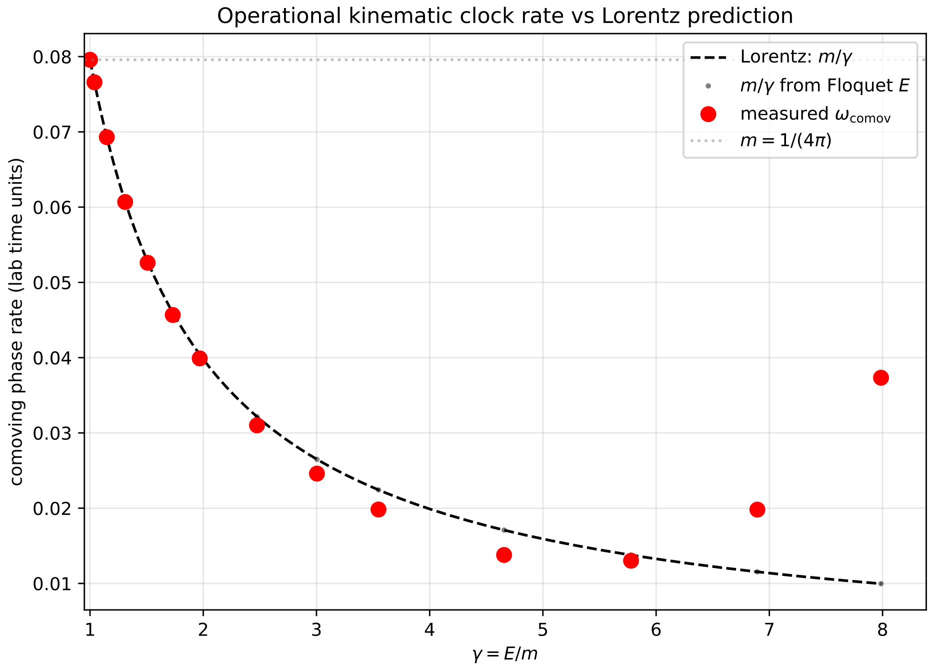

Figure 2: The operationally measured comoving phase rate ω_{comov} (red points) at moving wavepackets matches the Lorentz prediction m/γ (dashed line) across the range tested. The Floquet m/γ values (black dots) sit on the dashed line by construction. The horizontal grey line marks the rest mass $m = 1/(4\pi)$. Deviations at $\gamma > 3$ are the lattice's UV correction to Lorentz, quantified in Fig. 3.

7 Lattice corrections to Lorentz

The deviations of Table 3 are not numerical: they follow a smooth functional dependence and are reproduced when the run parameters $(L, \sigma, n_{\text{cycles}})$ are varied, provided no boundary wraparound contaminates the trajectory. They are the lattice's UV signature.

The leading correction can be read off the dispersion. Recall $\sin\delta = \delta - \delta^3/6 + O(\delta^5)$ and $\cos\delta = 1-\delta^2/2+\delta^4/24+O(\delta^6)$. Substituting into H_A+H_B at $k = (\pi-\delta, 0)$ yields a fourth-order correction to the dispersion,

$$\varepsilon^2(\delta) \;=\; m^2 + \tfrac{1}{4}\delta^2 - \tfrac{1}{12}\delta^4 + O(\delta^6), \tag{19}$$

which softens the Lorentz form at large δ. The corresponding correction to the comoving phase rate is

$$\omega_{\text{comov}} \;=\; \frac{m^2}{\varepsilon} \;\approx\; \frac{m}{\gamma}\left(1 - \frac{1}{12}\,\frac{\delta^4}{m^2+\delta^2/4} + O(\delta^6)\right). \tag{20}$$

At fixed γ, the lattice correction grows as δ^4: quartic in the momentum offset. The structure is generic to 2D nearest-neighbour Floquet rules with sin/cos hopping.

Statement. The lattice correction grows with momentum offset and becomes significant as the packet approaches the lattice cutoff. The numerical signature is that the operational comoving rate ω_{comov} falls below the Lorentz prediction m/γ at large γ, with a smooth functional dependence determined by the lattice dispersion. We do not assign a physical cutoff scale or convert the deviation to physical units in this paper; the deviation is a structural feature of the discrete substrate and would have to be matched against an external physical scale to become a quantitative falsifier.

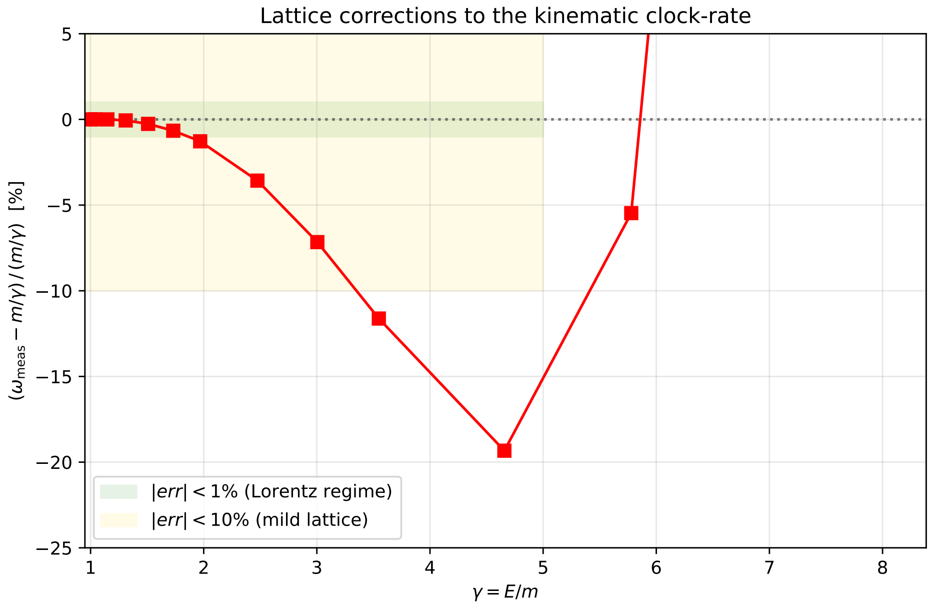

Figure 3: Relative deviation of the measured comoving clock rate from the Lorentz prediction m/γ, as a function of γ. The Lorentz regime (green band, $|err| < 1\%$) extends to $\gamma \approx 2$. Mild lattice corrections (gold band, $|err| < 10\%$) cover up to $\gamma \approx 4$. Beyond, the deviation grows monotonically and is the structural prediction of Eq. (20).

8 Bridge to Paper II

The two factors of weak-field general relativity — the gravitational time dilation $\sqrt{1 - 2GM/c^2 r}$ and the kinematic time dilation $\sqrt{1 - v^2/c^2}$ — are derived in DDD by structurally separate mechanisms:

- The gravitational factor (Paper II) comes from the *scalar* reserve sector with the source-side bandwidth mobility $\mu(R) = R/R_0$. The bandwidth at a node decreases near a sink, lowering the local rate of effective proper time to R/R_0.
- The kinematic factor (this paper) comes from the *spinor* sector with the Floquet two-step rule. The proper-time rate of a moving wavepacket is m/γ, where $\gamma = E/m$ is fixed by the Lorentz dispersion of the gap.

The combined factor for a moving spinor excitation in a gravitational background — a single-throttle scalar field $R(r) = R_0\sqrt{1 - \beta/r}$ acting on a propagating spinor ψ — is, schematically,

$$\frac{d\tau}{dt} \stackrel{?}{=} \frac{R}{R_0} \cdot \frac{m}{\gamma} = \sqrt{1 - \beta/r} \cdot \frac{1}{\gamma}, \tag{21}$$

under the conjecture that the two factors multiply rather than combine through a different scheme. The product form (21) is the same first-order combination as the weak-field general-relativistic combined factor, expressed as a product rather than as $\sqrt{1 - 2GM/c^2 r - v^2/c^2}$, with which it agrees to leading order in both β/r and v^2 but differs at second order. This is a structurally distinguishing prediction; we do *not* verify it numerically here, because we have not yet defined the coupling between the scalar and spinor sectors. The verification is deferred to Paper VII (coupling) and Paper IV (gravitational phenomenology beyond g_{00}).

9 What this paper does not do

We do not derive the Lorentz group, special relativity, or any covariance result. We derive a kinematic time-dilation factor m/γ as a continuum limit of a specific lattice rule, near a band gap, conditional on the Floquet two-step structure adopted in Sec. 3.

We do not derive the value of $\lambda = 1/(2\pi)$. It is a structural input fixed by the topology of the lattice (one diagonal twist link per plaquette diagonal, normalised to a single 2π winding per unit area). Whether this is uniquely selected by deeper consistency requirements is an open problem.

We do not derive the spinor sector itself. The two-component $\mathbb{C}^2$ field ψ_i at each node is adopted by analogy with Weyl semimetals [3, 4] and is the minimal non-trivial extension of the scalar substrate of Paper I that supports propagating excitations with a band gap. The motivation for this specific structure — as opposed to, say, Wilson [1] or Kogut–Susskind [2] fermions — is heuristic; alternative structures may produce equivalent low-energy effective theories.

We do not derive Lorentz invariance *globally*. The dispersion is Lorentz-like only near the gap point $(\pi, 0)$. Other gap points (none in this 2D model, but several in higher dimensions) would carry their own effective masses and velocities. The substrate is not Lorentz-invariant as a whole; it is Lorentz-invariant in the low-energy effective theory at each gap.

We do not address the photon-like massless sector. The Floquet rule has no gapless mode in this 2D version; gapless propagating modes require a different gap structure or a different lattice topology, and are deferred to Paper VI (gauge sector). The gravitational-propagation consequences of photon-like modes (deflection, Shapiro delay, spatial metric) are deferred to Paper IV.

We do not address coupling between the spinor and scalar sectors. The scalar reserve R is uniform throughout the spinor's evolution in this paper. The combined gravitational-kinematic factor of Eq. (21) is conjectural pending Paper VII.

We do not derive Newton's G, the speed of light c, or the mass scale m in physical units. The lattice quantities are $\lambda = 1/(2\pi)$, $T = 2$ (in lattice units), giving $m = 1/(4\pi)$ and $c_{\text{eff}} = 1/2$ (in inverse lattice units). Conversion to physical units requires the same external Planck-scale anchoring as in Paper II's *Structural form of Newton's constant* section.

10 Open questions

Origin of $\lambda = 1/(2\pi)$. The diagonal twist density is one 2π phase per plaquette diagonal. We adopt this by structural fiat. A microscopic argument that selects this density — perhaps from a topological invariant of the lattice or a flux-quantisation condition — would close the main remaining gap of this construction. The constant λ enters m and through it γ at every v, so its origin is structurally important.

Coupling to the scalar sector. In a static gravitational background $R(r) < R_0$, does the spinor's effective mass m depend on R? Does the effective c_{eff}? The natural ansatz is $m_{\text{eff}}(r) = m\,(R/R_0)$ or similar, but this is not derived. Paper VII addresses this.

Lattice corrections at high γ. The functional form (20) is the leading quartic correction. Higher-order corrections, and the question of whether the deviation continues to be monotonically negative at all γ or saturates, are open. A scan to $\gamma \sim 30$ on $L = 512$ would settle the issue numerically.

Massless modes. In the present 2D model the spinor sector has no gapless mode; all bands are gapped. A 3D extension of the Floquet rule, as explored in the Weyl-semimetal analogy [4], exhibits Weyl points where the gap closes. These are the candidates for photon-like massless excitations and are the subject of Paper VI.

11 Code and data availability

The Python scripts in `code/` implement the Floquet two-step rule of Eq. (5) on a 2D periodic lattice. They are pure-numpy and depend on no external simulation framework.

- `01_wavepacket_propagation.py`: verifies that wavepackets propagate at the predicted group velocity.
- `02_floquet_dispersion.py`: computes the Floquet quasi-energy spectrum and verifies the gap structure.
- `04_kspace_projection.py`: tracks the lab-frame energy of a wavepacket via $\langle k_0|\psi(t)\rangle$.
- `05_comoving_clock_rate.py`: implements the Galilean-shifted dealiased projection of Sec. 5 and produces Table 3.

The numerical data are stored in `data/`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-3-Inertia.

12 Outlook

This paper extracts the kinematic time-dilation factor of special relativity from a specific lattice rule near a band gap, as the operational analogue of the bandwidth-clock identification of Paper II. The result and the rule are conditional on the two-step Floquet structure adopted in Sec. 3; that structure has its own motivation and its own open questions, which we do not pretend to settle here.

1. Paper IV — predictions and tests of the gravitational sector beyond the static spherically symmetric scenario: photon deflection, spatial metric, compact-source phenomenology. *Mixed.*
2. Paper V — gravitational predictions and falsifiers: where DDD differs from Newton/GR and how to test it. *Mixed.*
3. Paper VI — the gauge sector and emergent electromagnetism, in particular the gapless modes of the Floquet rule in 3D and the connection to Maxwell. *Speculative; coefficients uncalibrated.*
4. Paper VII — coupling between the scalar and spinor sectors. Combined static-gravity $\times$ kinematic factor; verification of Eq. (21). *Conditional on Paper VI's gauge formulation.*

The result of this paper, taken on its own, is one structural fact: a discrete lattice rule with a Floquet two-step spinor sector produces, in the continuum limit near a band gap, the Lorentz time-dilation factor as the operational rate of effective proper time at moving excitations. The interpretation is intrinsic to the rule; no Lorentz invariance, no metric, and no Lorentzian proper-time formula is assumed; what is identified is the operational substrate quantity, not a metric postulate. The factor that emerges is the same as in special relativity, accompanied by lattice corrections at high γ that are calculable and, in principle, falsifiable.

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